Akshita Rao

Embedded Firmware Engineer

RESUME_1578

Location: Gurugram, Haryana, India | Email: akshita.rao.resume_1578@example.com | Phone: +91-00000-00000 (placeholder)
Total Experience: 5 years

PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 5 years of experience in embedded firmware, using C, C++, ARM Cortex-M, FreeRTOS. Experienced in real-time systems, hardware debugging and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: UART, Firmware Debugging, STM32, C++, C, Oscilloscope, FreeRTOS, SPI, JTAG, Embedded Linux, ARM Cortex-M, CAN

Methods: Real-time systems, Hardware debugging, Protocol validation

PROFESSIONAL EXPERIENCE

Senior Embedded Firmware Engineer | DeltaPoint Innovations | Jan 2023 - Present

- Lowered field-reset incidents by 36% with watchdog recovery and diagnostics
- Improved battery runtime by 16% with peripheral and sleep-state optimization
- Validated 630 hardware test vectors across protocol and timing corner cases
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Associate Embedded Firmware Engineer | Foresight Industrial | Jul 2021 - Dec 2022

- Validated 381 hardware test vectors across protocol and timing corner cases
- Reduced embedded task jitter by 15% using interrupt priority and queue tuning

PROJECTS

Motor Control Interface | FreeRTOS, ARM Cortex-M, CAN, C

- Implemented CAN message handling, fault logging and deterministic state transitions.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

Bootloader Update Utility | Firmware Debugging, Oscilloscope, CAN, FreeRTOS

- Developed verified firmware transfer and rollback checks.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

EDUCATION

B.Tech in Electronics and Communication Engineering, Shivanta College of Technology, 2021
Relevant Methods: Real-time systems, Hardware debugging, Protocol validation

CERTIFICATIONS

RTOS Scheduling Training (2021)